

CSE260

Assignment 03

Total Marks: 5X6=30

This assignment must be hand-written. Show ALL steps in ALL questions.

Q1

Build a circuit that implements the 2's complement number system (4 bit) using encoder(s) and decoder(s).

Q2

Implement the following boolean function using only one 2:1 mux

$$F(A,B,C,D) = \sum(0,1,5,7,8,11,12,13, 14)$$

Q3

Implement the following boolean function using 2x4 decoder(s) only

$$F(A,B,C,D,E) = \sum(0,2,5,9,15,22,13,18,21,28)$$

Q4

Design $A+2$ using a 4 bit parallel adder. Use external gates if required. The internal diagram for the parallel adder must be drawn using Encoders and decoders.

NB: Inside the parallel adder, the components you see are basic full adders. For a 4 bit parallel adder, you need 4 full adders. Those full adders must be built using encoder(s) and decoder(s).

Q5

You want to build a calculator that can add or subtract two 4-bit numbers: X and Y. Now, if X is greater than 9, and Y is divisible by 4, then the calculator will do the operation $X-Y$ (addition). Otherwise, the calculator will do the operation $X+Y$ (subtraction). Design the circuit using a 4-bit parallel adder cum subtractor.

Q6

Design a 10 person voter counting system using full and parallel adders. You must use encoder and decoder to build full adders, meaning you can't use full adders directly.